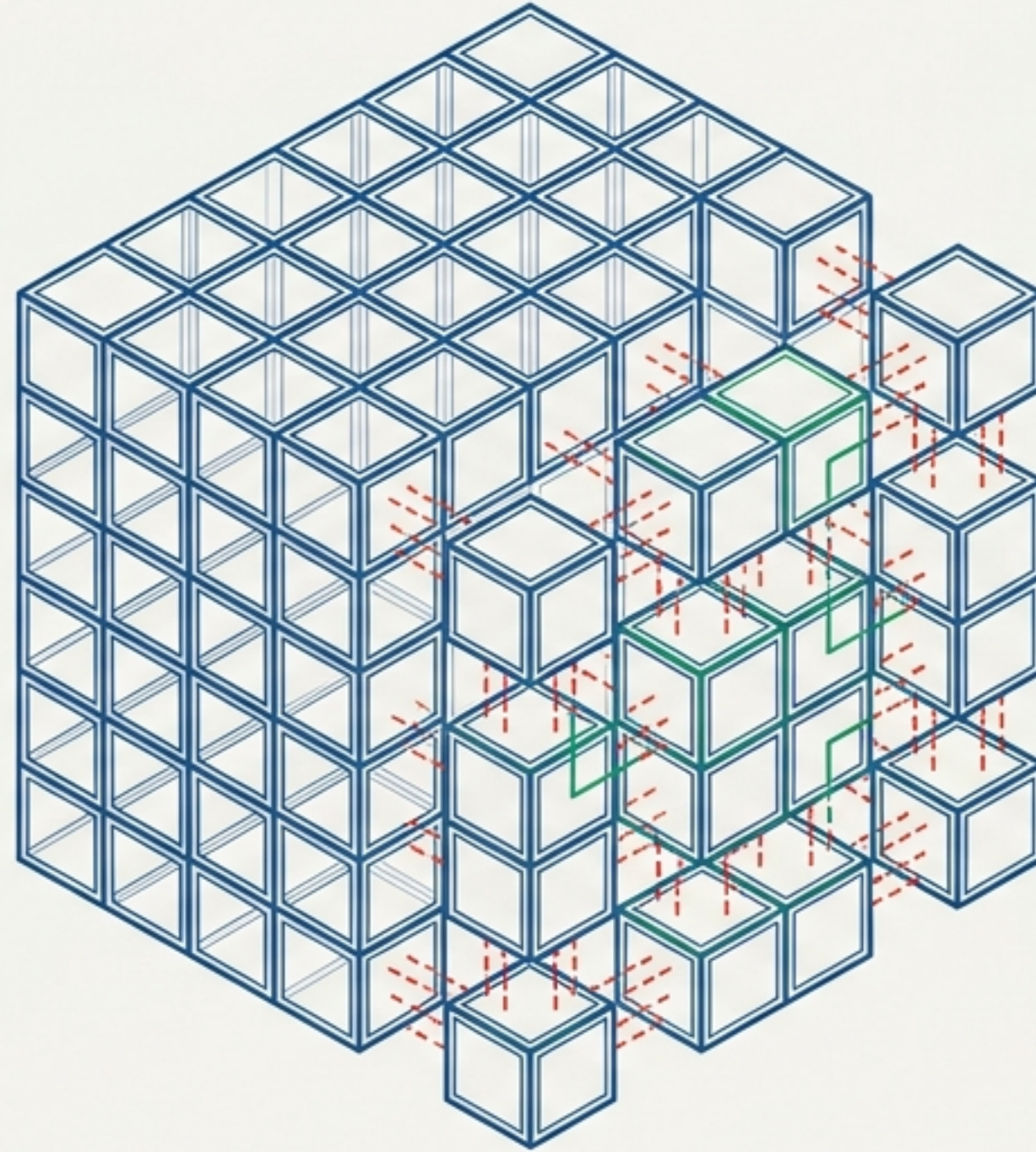


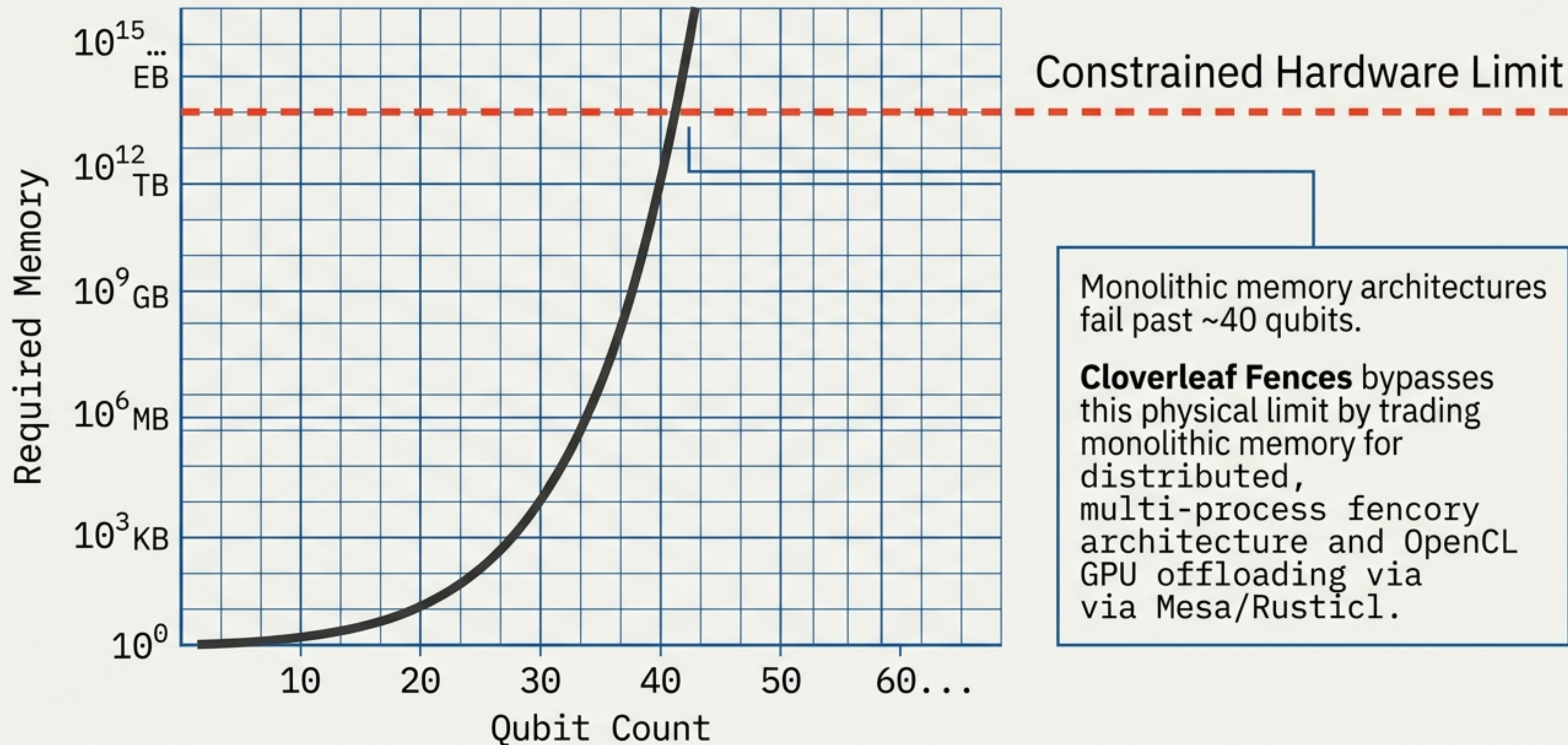
Cloverleaf Fences

Escaping the Quantum Memory Wall via Holographic Tensor Networks & Macroscopic Grid Annealing

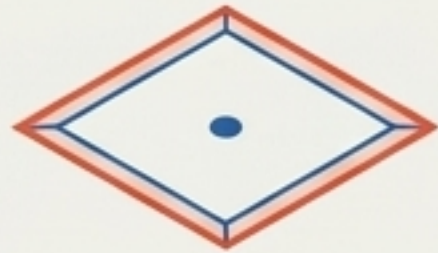


github.com/twobombs/thereminq-examples/cloverleaf-fences

The Exponential Constraint of Exact Simulation



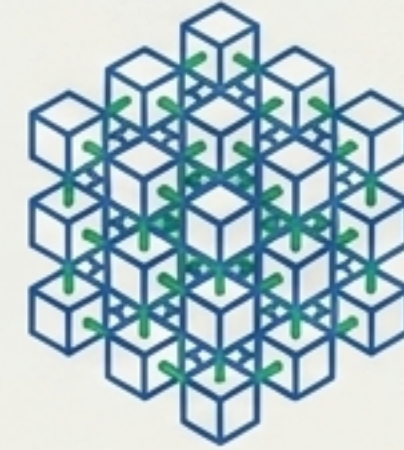
Three Engines to Bypass the Memory Wall



concept/



syk-rs/



lhv-ace-xyz/

Scale:
72-108 Qubits

Methodology:
Holographic Embedding

Inter-Patch Strategy:
Boundary Parameter Training

Scale:
144-300 Qubits

Methodology:
Traversable Wormholes / RCS

Inter-Patch Strategy:
Asynchronous IPC & Z-Kicks

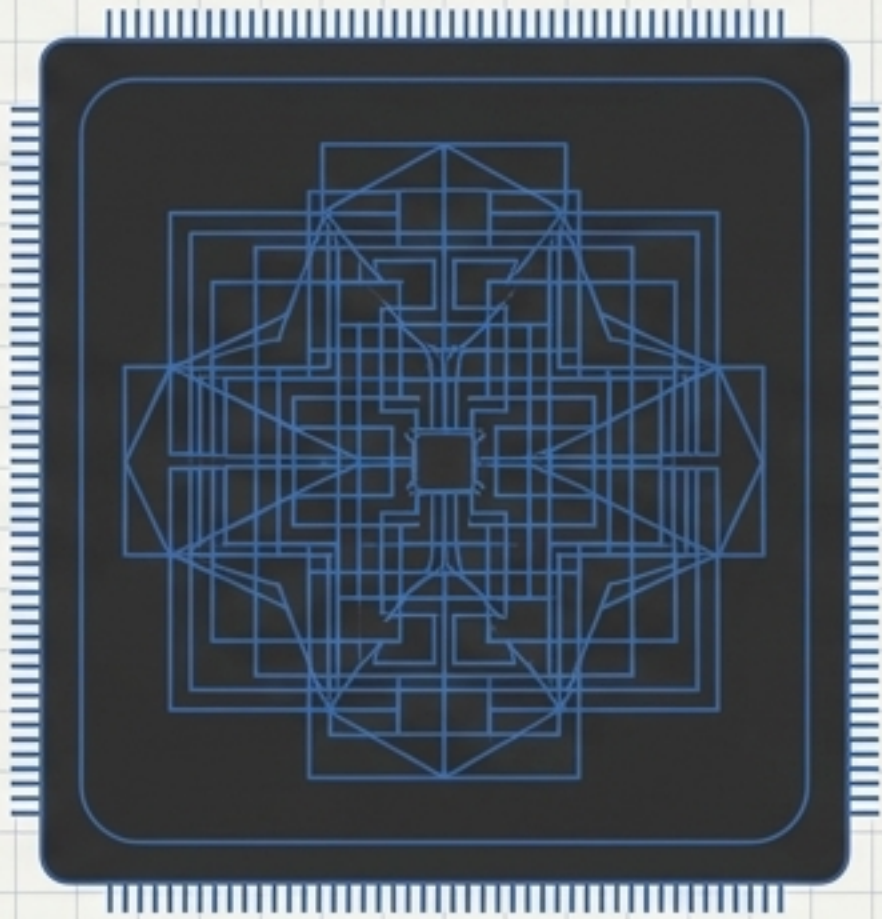
Scale:
Up to 729 Qubits

Methodology:
Macroscopic Grid Annealing

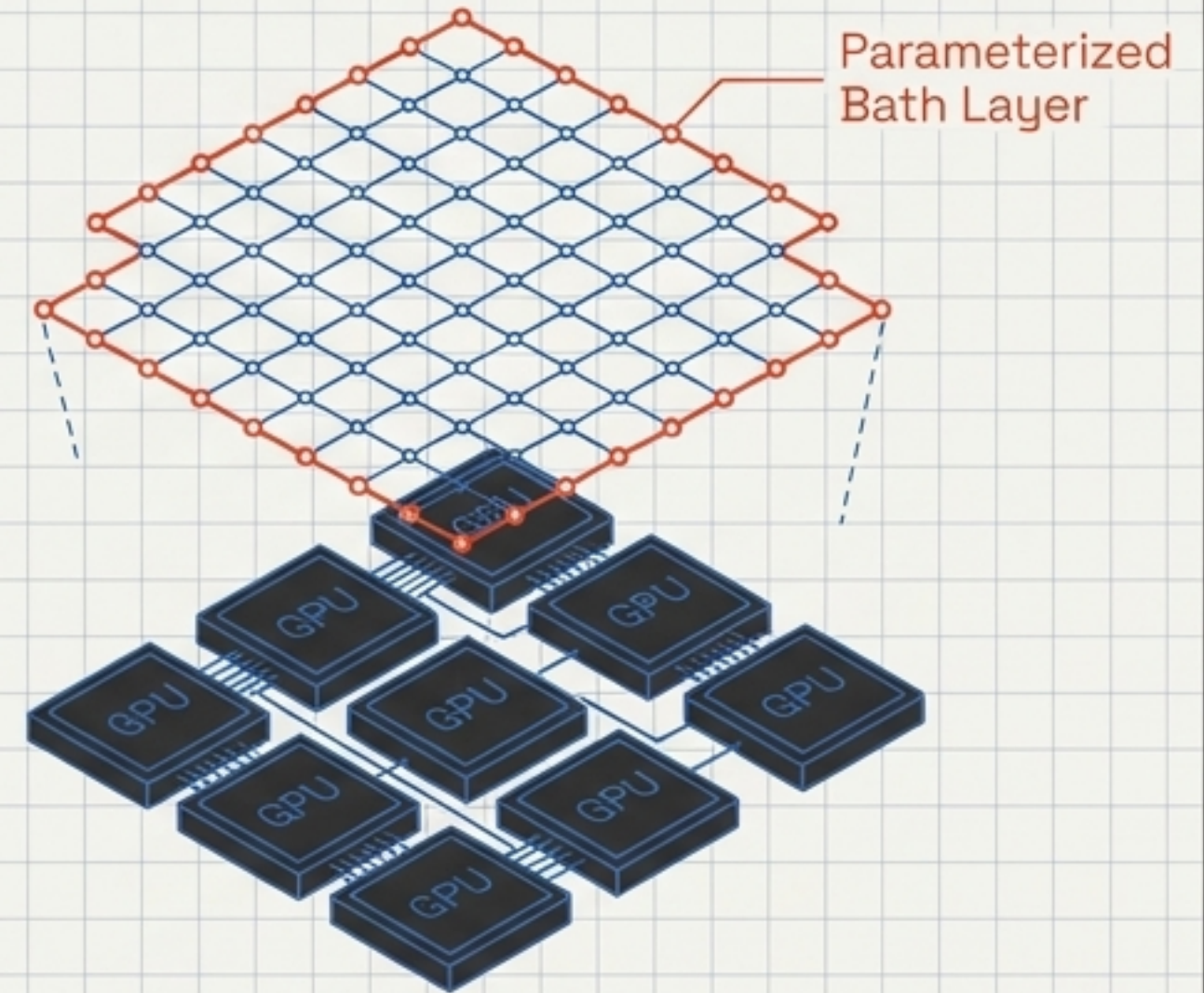
Inter-Patch Strategy:
Covariance Kicks &
Stochastic Variance

Reconstructing the Bulk from the Boundary

Monolithic CPU Oracle



Holographic Distributed Engine

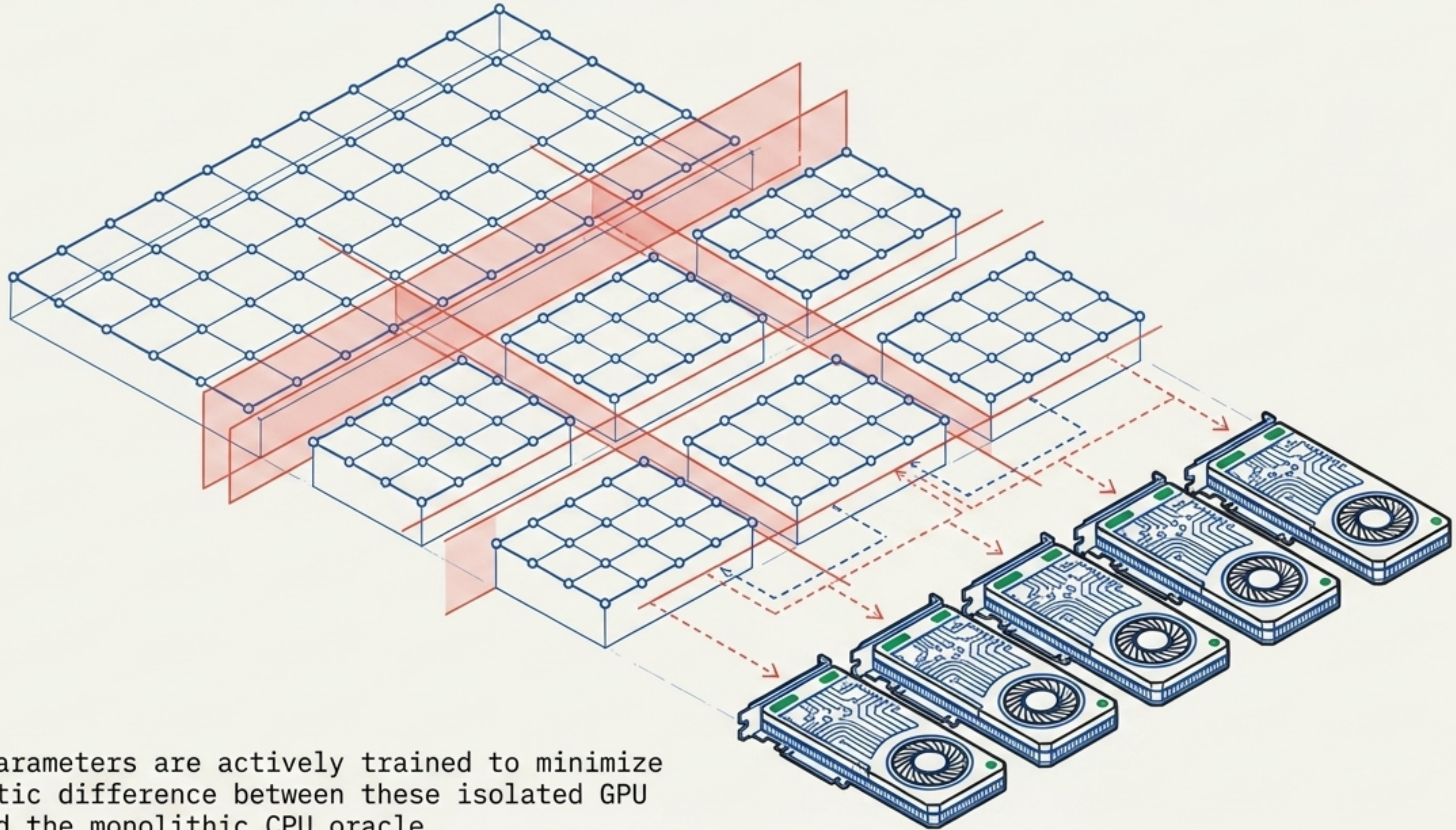


Evaluates exact ground truth energy via **PyQrack Binary Decision Tree (BDD)** across multiple processes.

Calculates Entanglement Embedding Loss. True cross-boundary entanglement is substituted with a parameterized bath layer.

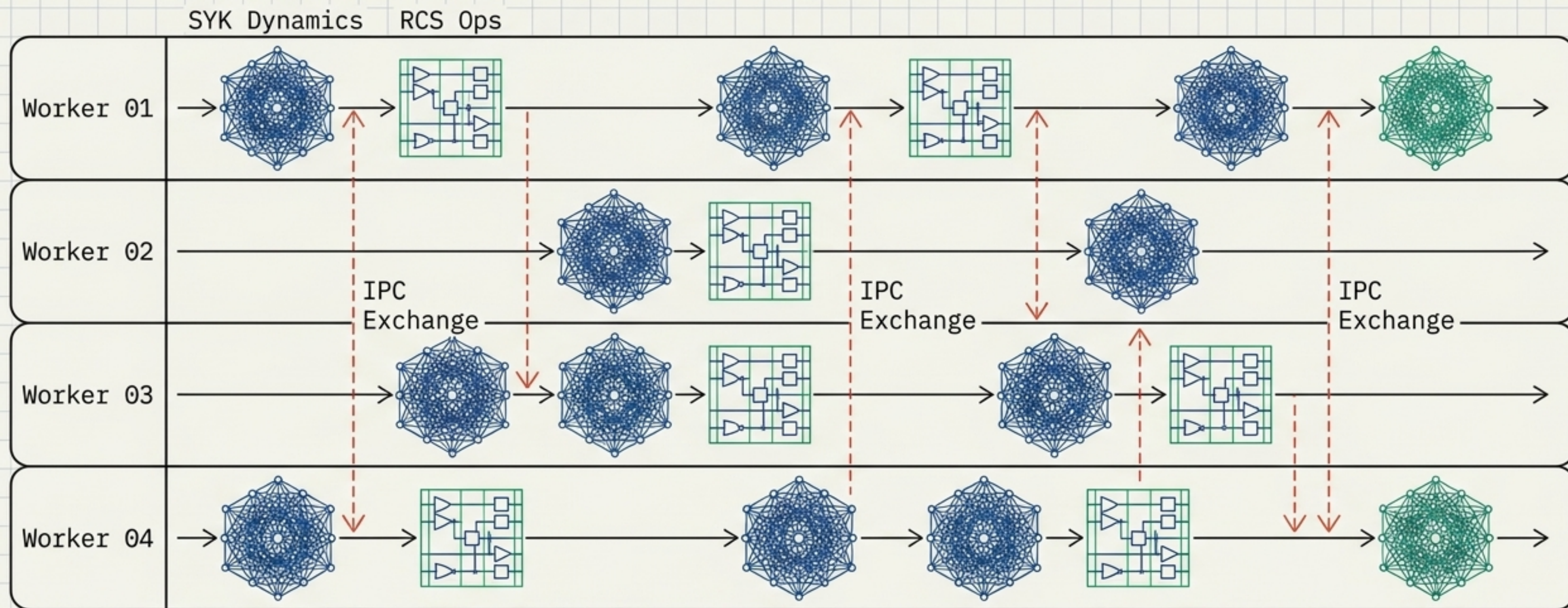
Fracturing the Global Grid (108 Qubits)

108-3x6x6-primary_concept.py



Boundary parameters are actively trained to minimize the energetic difference between these isolated GPU patches and the monolithic CPU oracle.

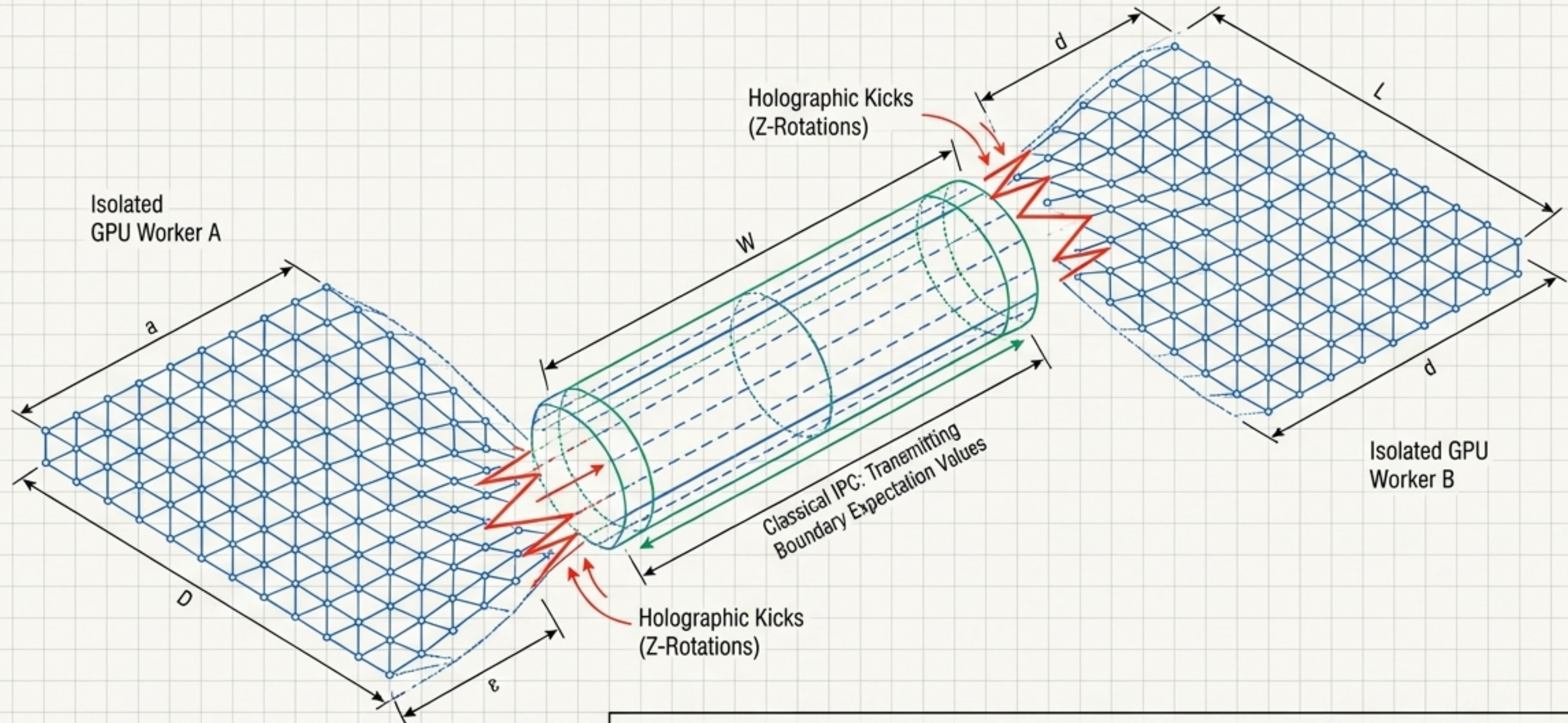
Emulating Deep Entanglement via SYK Dynamics



Persistent GPU workers execute in strict isolation but are orchestrated by a classical **IPC protocol** transmitting boundary **expectation values** in real-time.

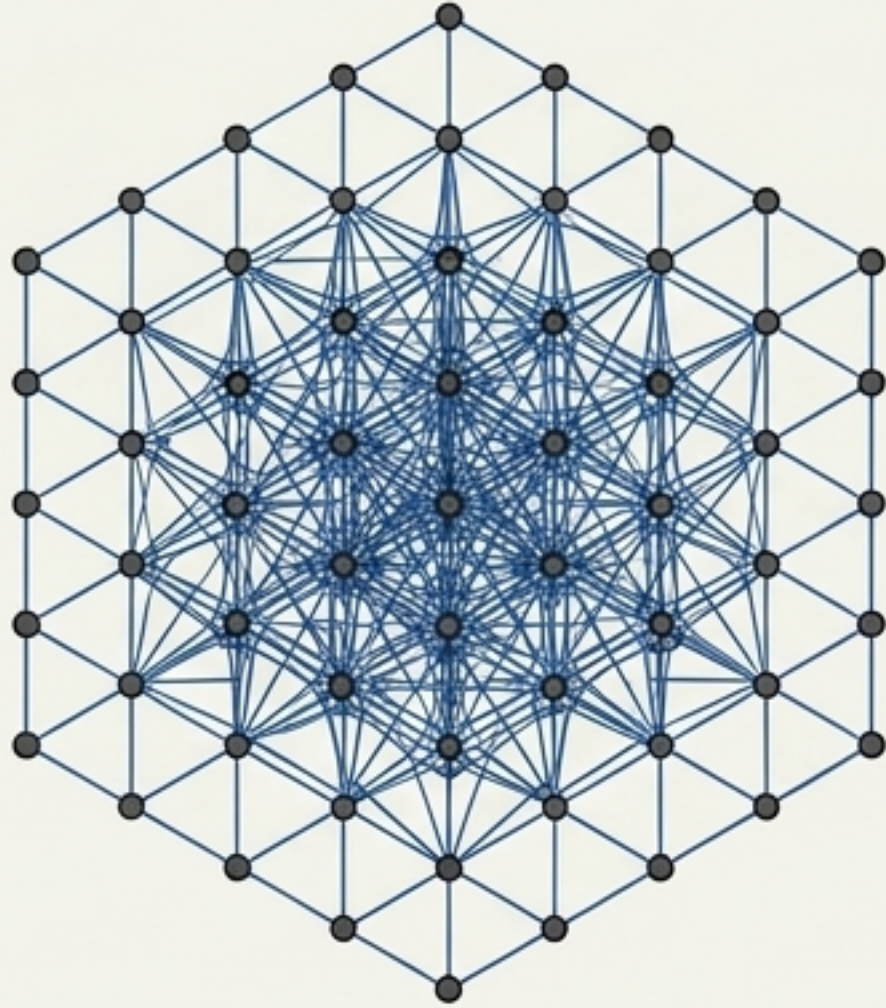
Engineering a Traversable Wormhole in Software

`traversable_wormhole_construct.py`



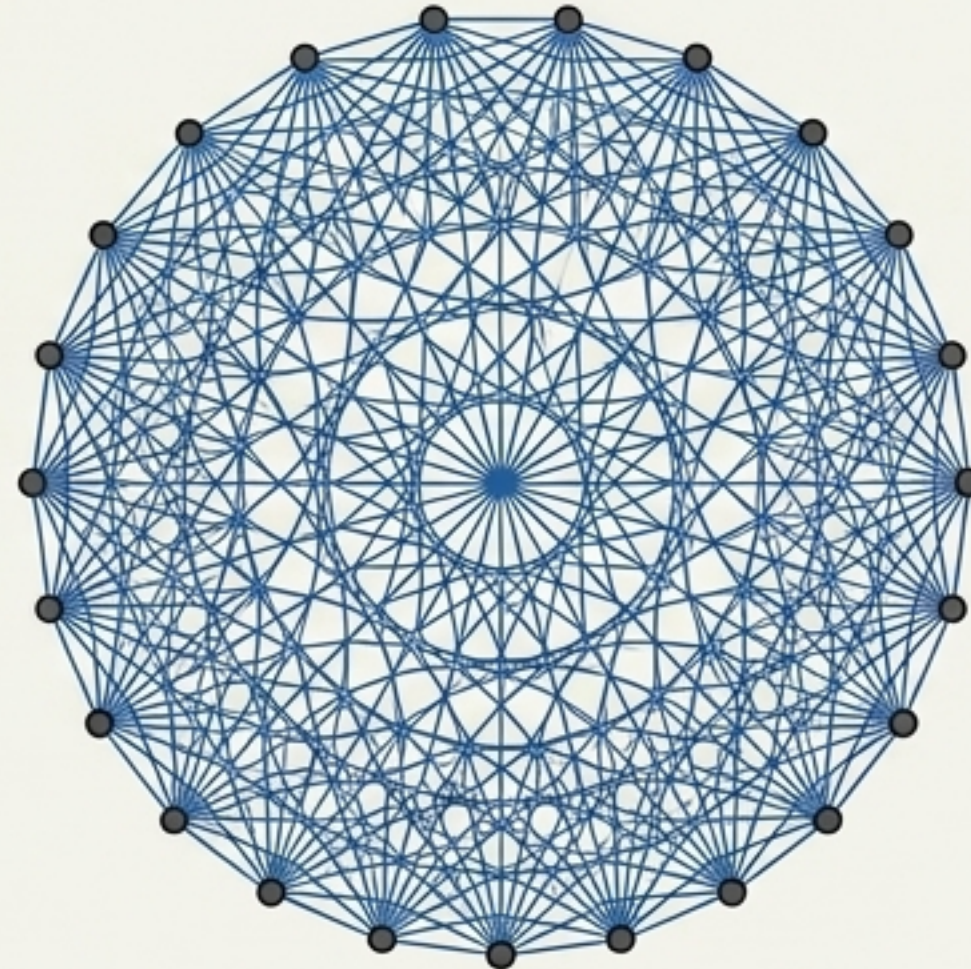
Holographic kicks inject negative energy shockwaves into the system, allowing state information to successfully traverse the simulated manifold between isolated patches.

Topologies for Complex Scrambling



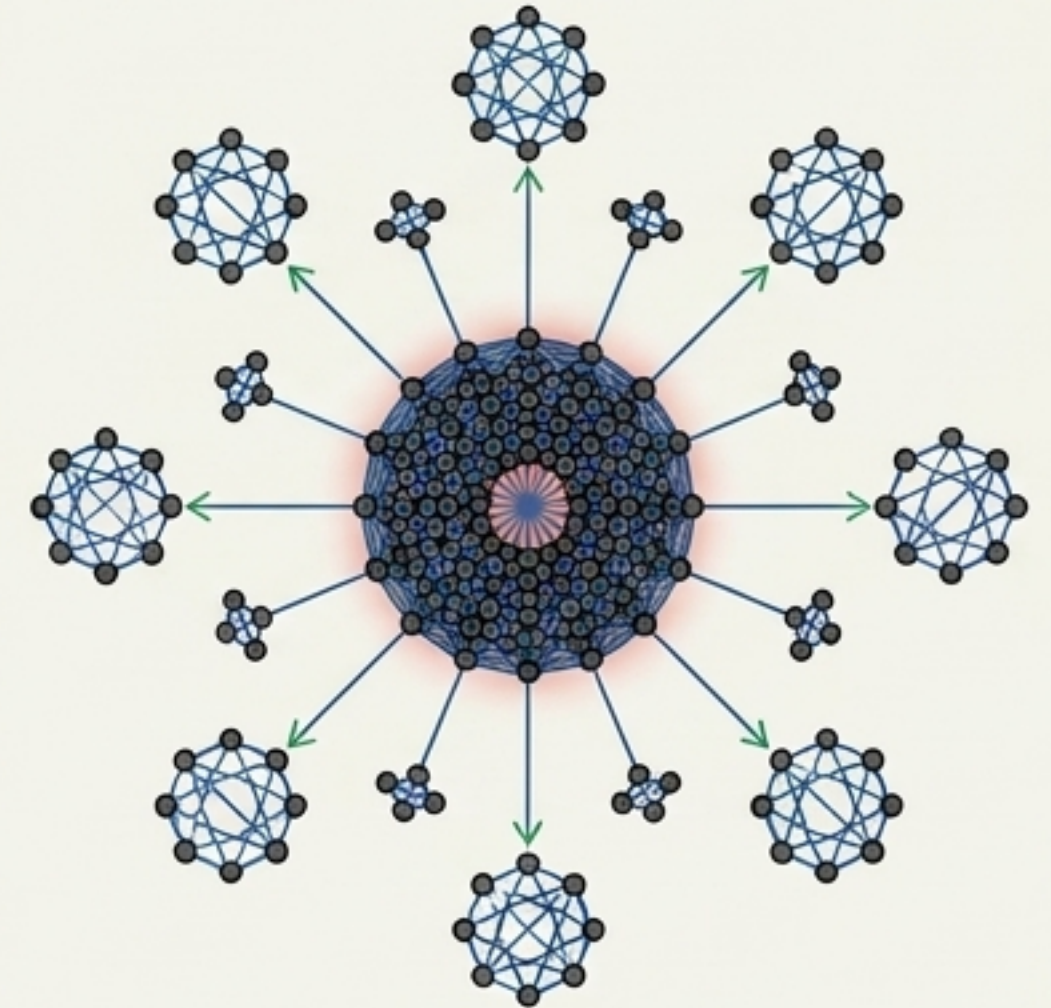
Nearest Neighbor (NN)

144q/216q dynamically fractured 3D lattices enabling traversable wormholes.



Complete Graph (FC)

Dense fully-connected meshes allowing cross-patch kicks and massive cross-correlations.



Unimatrix

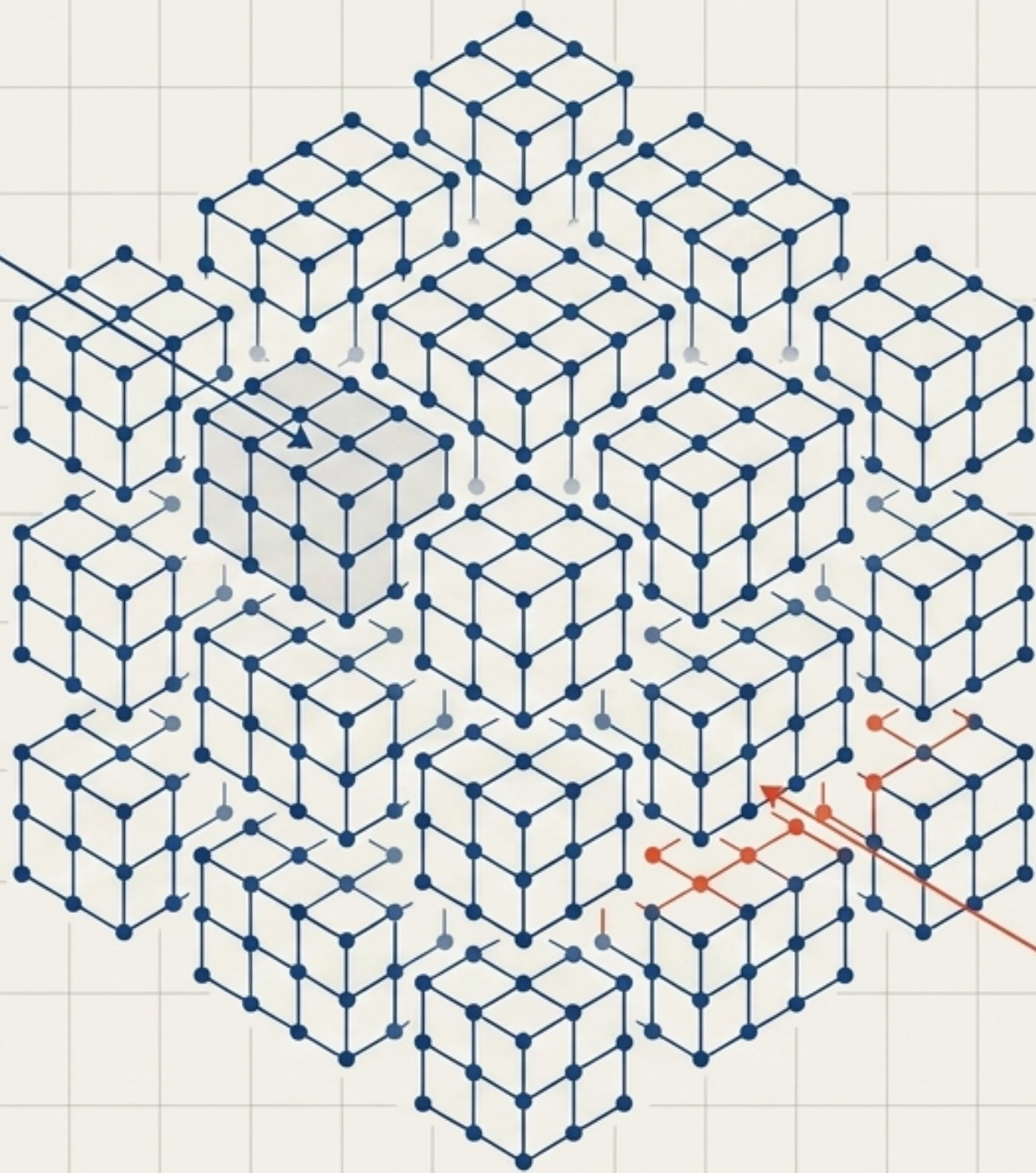
Centralized hub geometry utilizing mean-field holographic decoupling.

Scaling to Macroscopic Lattices (729 Qubits)

729_qubit_lattice_architecture.py

Intra-Patch Dynamics

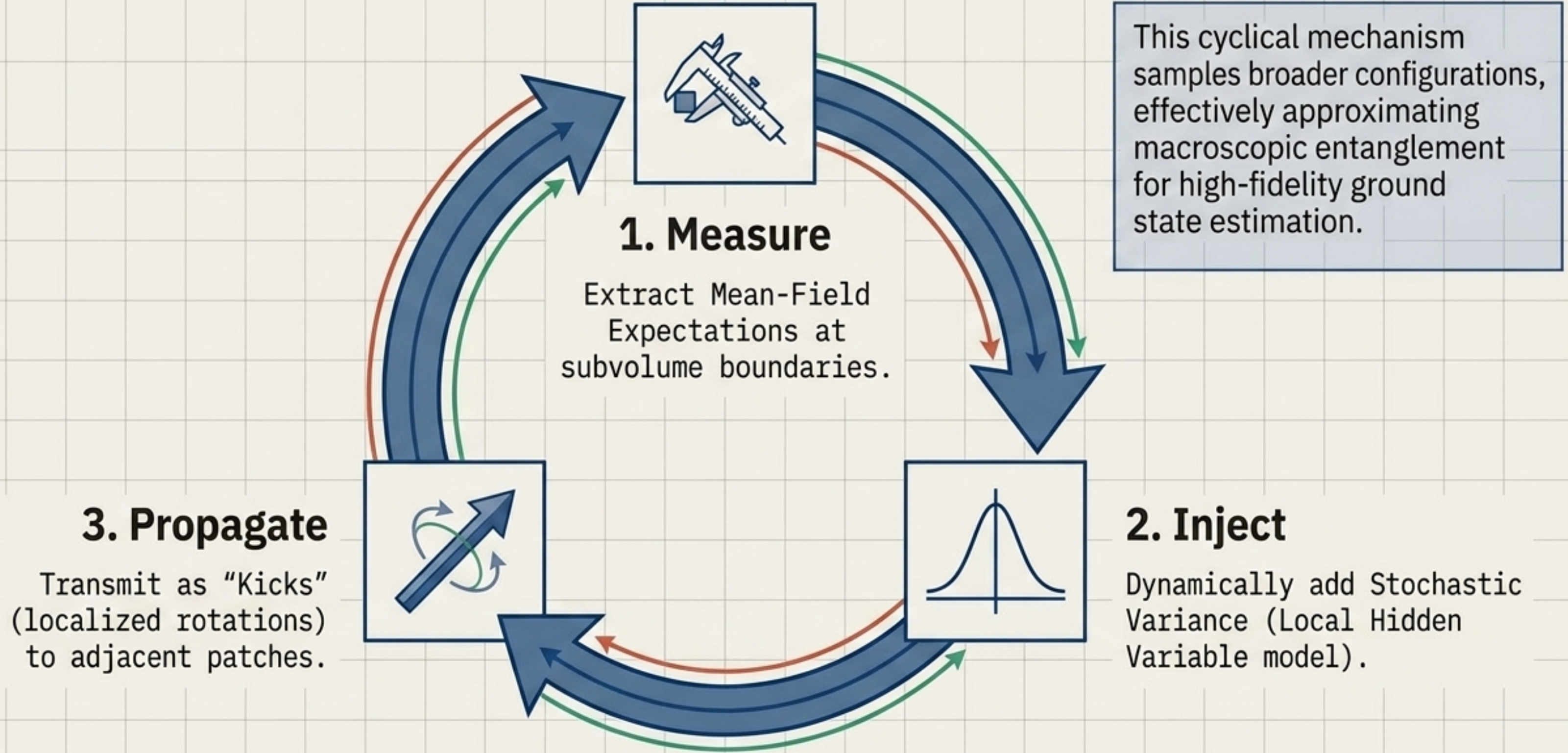
Governed by exact
Trotter-Suzuki
decomposition via GPU
statevector scaling.



Inter-Patch Correlations

Governed by mean-field
boundary interactions.

Approximating Entanglement with Covariance Kicks



Rooted in Foundational Quantum Theory

Foundational_Quantum_Theory.py

Theoretical Precedent

AdS/CFT & Holographic Principle (Swingle 2009)

Unitary Coupled Cluster & Fermionic Mappings (Romero 2017)

Circuit Cutting & Knitting (Bravyi 2021)

Cloverleaf Fences Architecture

Boundary bath layers in the Concept Engine

VQE Ansätze to Pauli string mappings

Fractured grids, Mean-Field Boundary Interactions & IPC

Enterprise-Scale Execution on Constrained Hardware

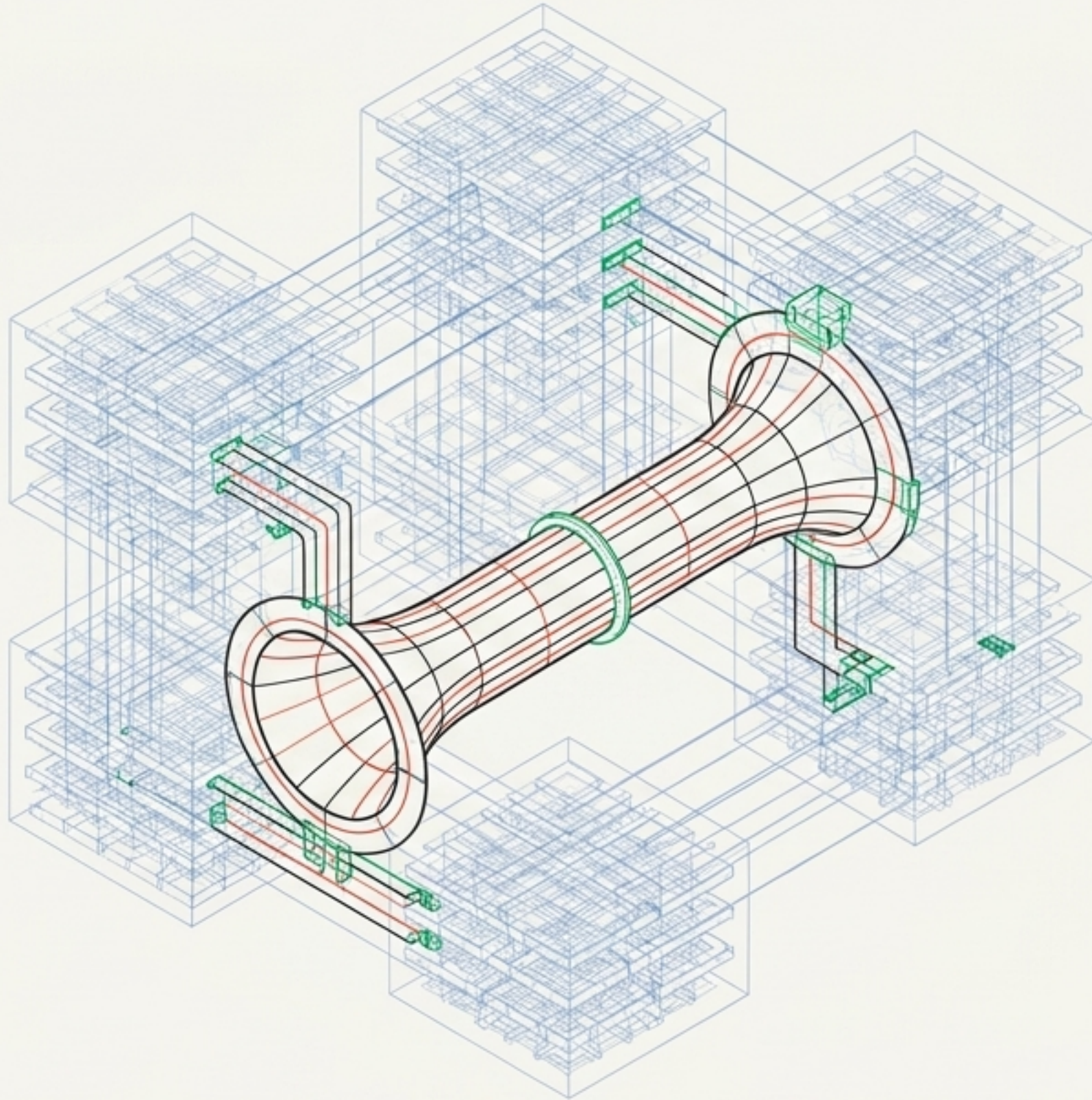
Maximizes memory bandwidth distribution across all NUMA nodes.

Deeply integrates OpenCL GPU offloading through Mesa/Rusticl.

```
user@numa-node:~/cloverleaf-fences$  
numactl --interleave=all  
python3 syk-rs/NN/144-3D-NN-syk-dev.py
```

Eliminates reliance on strict enterprise quantum hardware via distributed orchestration.

The Future of Distributed Quantum Simulation



Cloverleaf Fences successfully fractures the exponential memory wall, proving that classical orchestration, holographic theory, and strategic variance can emulate macroscopic quantum states on today's constrained hardware.

github.com/twobombs/thereminq-examples